

Sky Cover / Cloud Elements

The sky coverage, cloud base heights, and ceiling elements in GHCNv are equivalent to the legacy Global Hourly product known as the Integrated Surface Dataset (ISD). For those familiar with ISD, we have provided the matchup between GHCNv and ISD code groups of GA, GD, GF and the mandatory ceiling (CIG) element in the table below. The ISD manual can be found here: <https://www.ncei.noaa.gov/data/global-hourly/doc/isd-format-document.pdf>.

IMPORTANT UPDATE:

The previous version of GHCNv (v1.0.1) had only a limited number of sky elements, called sky_cover_[1-3] and sky_cover_baseht_[1-3]. This latest version (v1.1.0) dramatically expands the number of elements (see below). In addition, sky_cover_[1-3] and sky_cover_baseht_[1-3] are now called sky_cover_layer_[1-4] and sky_cover_layer_baseht_[1-4]. **EXCEPTION:** For **sources 343 and 344 only**, data from the previous elements, sky_cover_[1-3] and sky_cover_baseht_[1-3], are now in sky_cover_summation_[1-4] and sky_cover_summation_baseht_[1-4].

Depending on the data source, sky/cloud data are derived from Synoptic messages, METARs, SPECIs, Surface Airways Observations, among others.

SYNOP (Surface Synoptic Observations) are recorded in the sky_condition and sky_cover_layer elements. For the sky_cover_layer elements, each cloud layer is reported independently.

METARs (Meteorological Aerodrome Reports) and SAOs (Surface Airways Observations) are generally reported in the sky_cover_summation elements for the U.S./Canada and in sky_cover_layer elements for all other stations (international). The U.S./Canada stations follow the summation principle, which means that each reported cloud layer adds to the total sky coverage; whereas, International stations report each cloud layer independently (discrete layers).

Note: not all sky elements are reported for any given station.

Quality codes are source-specific and are listed in the [main GHCNv readme document](#). Source codes are found here: [ghcnv-source-list.pdf](#)

GHCNv elements and associated ISD code groups:

GHCNv element names	ISD code group
sky_condition	GF1 SKY SKY-CONDITION-OBSERVATION
sky_condition_baseht	GF1 SKY SKY-CONDITION-OBSERVATION
ceiling_height	SKY-CONDITION-OBSERVATION ceiling height (mandatory data section)
sky_cover_layer_1 - 4	GA1-GA4 SKY-COVER-LAYER
sky_cover_layer_baseht_1 -4	GA1-GA4 SKY-COVER-LAYER
sky_cover_summation_1 -4	GD1-GD4 SKY-COVER-SUMMATION-STATE
sky_cover_summation_baseht_1 -4	GD1-GD4 SKY-COVER-SUMMATION-STATE

sky_condition

Sky condition observation coverage codes for (1) total coverage code: denotes the fraction of the total celestial dome covered by clouds or other obscuring phenomena; (2) total opaque coverage code: denotes the fraction of the total celestial dome covered by opaque clouds or other obscuring phenomena (only reported by selected U.S. stations during selected periods); and (3) total lowest cloud coverage code: represents the fraction of the celestial dome covered by all low clouds present. If no low clouds are present; the code denotes the fraction covered by all middle level clouds present.

Data field: contains up to 3 coverage codes separated by a semicolon:

total ; opaque ; lowest

Examples:

00;;05
;;01
02;;

Coverage code key:

00 = None, SKC or CLR
01 = One okta - 1/10 or less but not zero
02 = Two oktas - 2/10 - 3/10, or FEW
03 = Three oktas - 4/10
04 = Four oktas - 5/10, or SCT
05 = Five oktas - 6/10
06 = Six oktas - 7/10 - 8/10
07 = Seven oktas - 9/10 or more but not 10/10, or BKN
08 = Eight oktas - 10/10, or OVC
09 = Sky obscured, or cloud amount cannot be estimated
10 = Partial obscuration
11 = Thin scattered
12 = Scattered
13 = Dark scattered
14 = Thin broken
15 = Broken
16 = Dark broken
17 = Thin overcast
18 = Overcast
19 = Dark overcast

Quality field: contains up to 2 quality codes for total and and lowest coverage codes separated by semicolon:

total ; lowest

Examples:

2;
2;1

;1

; (no codes reported)

*note: there is no quality code for opaque coverage codes

sky_condition_baseht

Height above ground level (AGL) of the base of the lowest cloud layer. Units: meters. Corresponds with lowest coverage code in sky_condition. Note: there are no base heights for the mid or high cloud layers given. The measurement code field contains cloud genus types for low, mid, and high cloud layers.

Measurement code: up to 3 cloud genus codes for low, mid and high cloud layers each separated by semicolon with corresponding quality code after the colon:

low : QC ; mid : QC ; high : QC

Examples:

08:1;04:9;;

;;07:1;;

;;;: (no codes reported)

Lowest Cloud Genus code:

00 = No low clouds

01 = Cumulus humilis or Cumulus fractus other than of bad weather or both

02 = Cumulus mediocris or congestus, with or without Cumulus of species fractus or humilis or Stratocumulus all having bases at the same level

03 = Cumulonimbus calvus, with or without Cumulus, Stratocumulus or Stratus

04 = Stratocumulus cumulogenitus

05 = Stratocumulus other than Stratocumulus cumulogenitus

06 = Stratus nebulosus or Stratus fractus other than of bad weather, or both

07 = Stratus fractus or Cumulus fractus of bad weather, both (pannus) usually below Altostratus or Nimbostratus.

08 = Cumulus and Stratocumulus other than Stratocumulus cumulogenitus, with bases at different levels

09 = Cumulonimbus capillatus (often with an anvil), with or without Cumulonimbus calvus, Cumulus, Stratocumulus, Stratus or pannus

Middle Cloud Genus codes:

00 = No middle clouds

01 = Altostratus translucidus

02 = Altostratus opacus or Nimbostratus

03 = Altocumulus translucidus at a single level

04 = Patches (often lenticulre) of Altocumulus translucidus, continually changing and occurring at one or more levels

05 = Altocumulus translucidus in bands, or one or more layers of Altocumulus translucidus or opacus, progressing invading the sky; these Altocumulus clouds generally thicken as a whole

- 06 = Altocumulus cumulogenitus (or cumulonimbogenitus)
- 07 = Altocumulus translucidus or opacus in two or more layers, or Altocumulus opacus in a single layer, not progressively invading the sky, or Altocumulus with Altostratus or Nimbostratus
- 08 = Altocumulus castellanus or floccus
- 09 = Altocumulus of a chaotic sky; generally at several levels

High Cloud Genus codes:

- 00 = No High Clouds
- 01 = Cirrus fibratus, sometimes uncinus, not progressively invading the sky
- 02 = Cirrus spissatus, in patches or entangled sheaves, which usually do not increase and sometimes seem to be the remains of the upper part of a Cumulonimbus; or Cirrus castellanus or floccus
- 03 = Cirrus spissatus cumulonimbogenitus
- 04 = Cirrus uncinus or fibratus, or both, progressively invading the sky; they generally thicken as a whole
- 05 = Cirrus (often in bands) and Cirrostratus, or Cirrostratus alone, progressively invading the sky; they generally thicken as a whole, but the continuous veil does not reach 45 degrees above the horizon
- 06 = Cirrus (often in bands) and Cirrostratus, or Cirrostratus alone, progressively invading the sky; they generally thicken as a whole; the continuous veil extends more than 45 degrees above the horizon, without the sky being totally covered.
- 07 = Cirrostratus covering the whole sky
- 08 = Cirrostratus not progressively invading the sky and not entirely covering it
- 09 = Cirrocumulus alone, or Cirrocumulus predominant among the High clouds

ceiling_height

Height above ground level (AGL) of the lowest cloud or obscuring phenomena layer aloft with 5/8 or more summation total sky cover, which may be predominantly opaque, or the vertical visibility into a surface-based obstruction. Units: meters. Unlimited ceiling is 22000 meters.

Measurement code: up to 2 codes separated by a semicolon: 1) the determination code that denotes the method used to determine the ceiling and 2) the CAVOK code which indicates whether the 'Ceiling and Visibility Okay' condition has been reported:

determination ; CAVOK

Examples:

C;
W;Y
;N
; (no codes reported)

Ceiling Determination code:

A = Aircraft

B = Balloon
C = Statistically derived
D = Persistent cirriform ceiling (pre-1950 data)
E = Estimated
M = Measured
P = Precipitation ceiling (pre-1950 data)
R = Radar
S = ASOS augmented
U = Unknown ceiling (pre-1950 data)
V = Variable ceiling (pre-1950 data)
W = Obscured

CAVOK code:

N = No
Y = Yes

sky_cover_layer_[1,2,3,4]

Up to 4 codes denoting the fraction of the total celestial dome covered by each sky cover layer. This is for *discrete cloud layers* (as opposed to the summation of cloud layers in the sky_cover_summation elements).

Data field: contains a string description (abbreviated) and its corresponding numeric code separated by colon:

sky cover abbreviation string : code

Example:

FEW:02

Coverage code key:

CLR:00 = None, SKC or CLR
FEW:01 = One okta - 1/10 or less but not zero
FEW:02 = Two oktas - 2/10 - 3/10, or FEW
SCT:03 = Three oktas - 4/10
SCT:04 = Four oktas - 5/10, or SCT
BKN:05 = Five oktas - 6/10
BKN:06 = Six oktas - 7/10 - 8/10
BKN:07 = Seven oktas - 9/10 or more but not 10/10, or BKN
OVC:08 = Eight oktas - 10/10, or OVC
VV:09 = Sky obscured, or cloud amount cannot be estimated
X:10 = Partial obscuration

sky_cover_layer_baseht_[1,2,3,4]

The height relative to a vertical-reference-datum of the lowest surface of a cloud. Units: meters. Up to 4 discrete cloud base heights at lowest point of layer; clear skies reported as a single layer; corresponds with coverage codes in sky_cover_layer_1-4 elements

Measurement code: contains the classification of the clouds that comprise a sky cover layer and the quality code for the reported cloud type separated by a colon with no space

cloud type code : quality code

Example:

00:5

: (no cloud type given by source)

Type codes:

00 = Cirrus (Ci)

01 = Cirrocumulus (Cc)

02 = Cirrostratus (Cs)

03 = Altocumulus (Ac)

04 = Altostratus (As)

05 = Nimbostratus (Ns)

06 = Stratocumulus (Sc)

07 = Stratus (St)

08 = Cumulus (Cu)

09 = Cumulonimbus (Cb)

10 = Cloud not visible owing to darkness, fog, duststorm, sandstorm, or other analogous phenomena / sky obscured

11 = Not used

12 = Towering Cumulus (Tcu)

13 = Stratus fractus (Stfra)

14 = Stratocumulus Lenticular (Scsl)

15 = Cumulus Fractus (Cufra)

16 = Cumulonimbus Mammatus (Cbmam)

17 = Altocumulus Lenticular (Acsl)

18 = Altocumulus Castellanus (Accas)

19 = Altocumulus Mammatus (Acmam)

20 = Cirrocumulus Lenticular (Ccsl)

21 = Cirrus and/or Cirrocumulus

22 = Stratus and/or Fracto-stratus

23 = Cumulus and/or Fracto-cumulus

sky_cover_summation_[1,2,3,4]

Up to 2 sky cover summation coverage codes can be reported for up to 4 vertical layers. Code 1 denotes the portion of the total celestial dome covered by all layers of clouds and other obscuring phenomena at or below a given height. Code 2 denotes the fraction of the total celestial dome covered by all layers of clouds and other

obscuring phenomena at or below a given height, if reported by the station in octas. Each reported cloud layer adds to the total sky coverage and is the *summation of cloud layers* (as opposed to the discrete layers found in the sky_cover_layer elements). Since up to 4 cloud layers can be reported, the full state of the sky can best be determined by the last layer's value. In other words if three layers are reported and the third layer uses BKN then the total state of sky is BKN which is similar in definition to “mostly cloudy.” OVC is similar to “cloudy” or overcast and FEW or SCT is similar to “partly cloudy.” These coverage codes correspond to the sky cover summation base height layers. The highest reported layer is the total sky coverage up to that level.

Data field: may contain one or both coverage codes: code 1 has an abbreviated string, then colon, then its corresponding numeric code (e.g., FEW:02) followed by semicolon and code 2 which is just the numeric code (e.g., 02).

sky cover code 1 string : sky cover code 1 ; sky cover 2 code

Examples:

FEW:02;02 (both codes reported)

;01 (no code 1 reported)

FEW:02; (no code 2 reported)

*note: only 1 quality code is given for both sky cover codes

Coverage code 1 key:

CLR:00 = Clear - No coverage

FEW:02 = FEW - 2/8 or less coverage (not including zero)

SCT:04 = SCATTERED - 3/8-4/8 coverage

BKN:07 = BROKEN - 5/8-7/8 coverage

OVC:08 = OVERCAST - 8/8 coverage

VV:09 = OBSCURED

X:10 = PARTIALLY OBSCURED

Coverage code 2 key:

00 = None, SKC or CLR

01 = One okta - 1/10 or less but not zero

02 = Two oktas - 2/10 - 3/10, or FEW

03 = Three oktas - 4/10

04 = Four oktas - 5/10, or SCT

05 = Five oktas - 6/10

06 = Six oktas - 7/10 - 8/10

07 = Seven oktas - 9/10 or more but not 10/10, or BKN

08 = Eight oktas - 10/10, or OVC

09 = Sky obscured, or cloud amount cannot be estimated

10 = Partial Obscuration

11 = Thin Scattered

12 = Scattered

13 = Dark Scattered

14 = Thin Broken

15 = Broken

16 = Dark Broken
17 = Thin Overcast
18 = Overcast
19 = Dark overcast

sky_cover_summation_baseht_[1-4]

Height above ground level (AGL) of the base of the cloud layer or obscuring phenomena. Units: meters. Up to 4 vertical layers can be reported. Corresponds with coverage codes in sky_cover_summation_1-4 elements. Each reported cloud layer adds to the total sky coverage and is the *summation of cloud layers* (as opposed to the discrete layers found in the sky_cover_layer_baseht elements).

Measurement code: a code that represents a characteristic of a specific cloud or other obscuring phenomena layer with no quality code. *NOTE: for sources 343 and 344, the measurement code indicates the cloud type and its QC code separated by a colon. When no code is given, the field will contain only a colon.

Examples:

09:5 (source codes 343/344)
: (no codes given)
2 (all other source codes)

Code key for most sources:

1 = Variable height
2 = Variable amount
3 = Thin clouds
4 = Dark layer (reported in data prior to 1950)

Code key for sources 343 and 344:

09 = Cumulonimbus (Cb)
12 = Towering Cumulus (Tcu)